Rajesh Phulwaria

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GitHub | LinkedIn | Portfolio

Profile

BCA student focused on machine learning and deep learning, with hands-on work in neural-network experimentation, Transformer components, retrieval-augmented generation (RAG), and API development. Interested in AI engineering and empirical research on neural-network training dynamics.

Technical Skills

Languages: Python, C++, JavaScript, SQL
ML / Deep Learning: PyTorch, TensorFlow, scikit-learn, CNNs, LSTMs, model evaluation
LLM Applications: Transformers, tokenization, embeddings, LangChain, vector search, RAG, LLM APIs
Backend / Infrastructure: FastAPI, REST APIs, Docker, AWS EC2, Linux, Git
Data / Research: NumPy, Pandas, Matplotlib, experiment design, literature review

Research

Controlled Gradient-Flow Ablation in MLPs Paper / DOI | Code

- Designed controlled experiments to study gradient-flow behaviour in multilayer perceptrons, examining how network depth, activation functions, and weight initialisation relate to gradient propagation.
- Used layer-wise gradient diagnostics and comparisons across experimental conditions to investigate factors associated with vanishing gradients.

A Comparative Analysis of Gradient Flow and Vanishing Gradient Effects in Deep Neural Networks Paper / DOI

- Examined the effect of diminishing gradients on backpropagation, parameter updates, and optimisation in deep neural networks, connecting theoretical explanations with training challenges.

Selected Projects

Triage – Explainable OPD Patient Prioritisation GitHub | Demo

- Built a healthcare workflow prototype that categorises OPD patients as Immediate, Soon, or Routine from symptoms and available vitals, with explanations for staff review.
- Implemented a staff dashboard, nurse override workflow, and audit trail in Next.js. Ranked in the top 15 of 72 teams at the AAYU MedTech hackathon.

Niyamit – Legal Metrology Compliance System GitHub

- Developing a packaged-commodity label verification system using OCR, NLP, structured extraction, and rule-based validation against India's Legal Metrology requirements.

Patient Data Management API GitHub

- Built CRUD endpoints and basic health-related calculations with FastAPI; used Docker, Streamlit, and AWS EC2 to separate the API and frontend and deploy the application.

YouTube Video Chatbot (RAG) Colab

- Built a question-answering pipeline over YouTube transcripts using LangChain, text chunking, embeddings, vector retrieval, and LLM-generated responses.

Transformer and GPT-Style Tokenizer GitHub

- Implemented language-model building blocks to explore BPE tokenization, embeddings, attention, and Transformer components.

Experience

Data Science Intern – Cognifyz Technologies *Machine Learning*

- Worked on ML workflows involving data preprocessing, feature engineering, experimentation, and model evaluation.

Python Programming Intern – InternsForge *Python Development*

- Developed Python applications and strengthened programming fundamentals and algorithmic problem-solving.

Education

Bachelor of Computer Applications (BCA) 2024–2027
Aishwarya College, Jodhpur, Rajasthan
Affiliated with Jai Narain Vyas University (J.N.V.U.)